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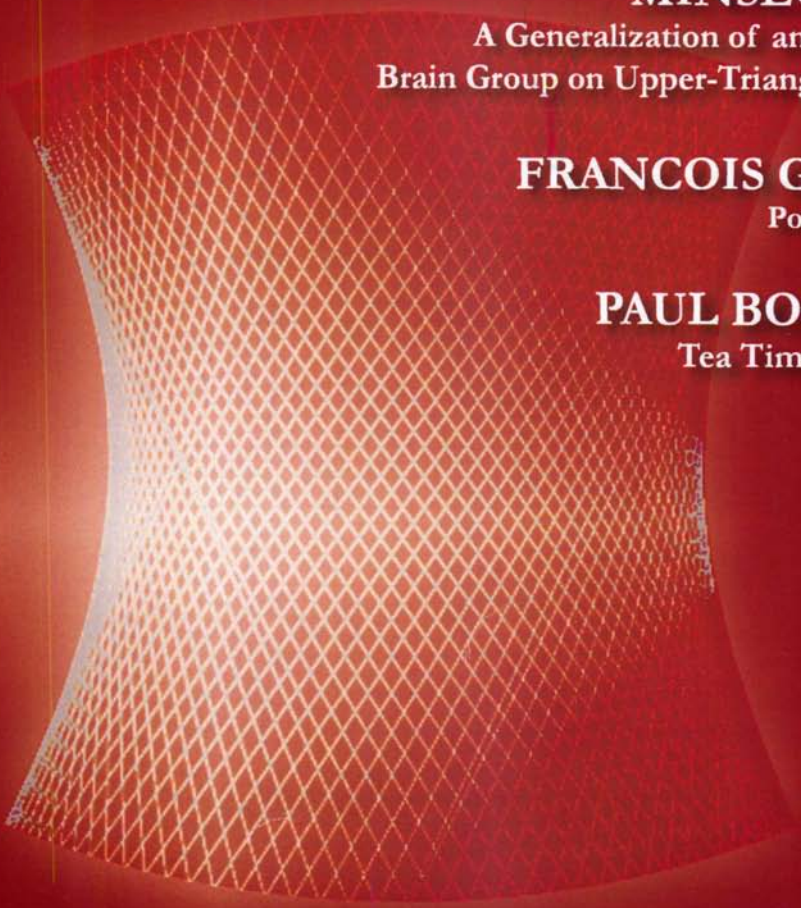

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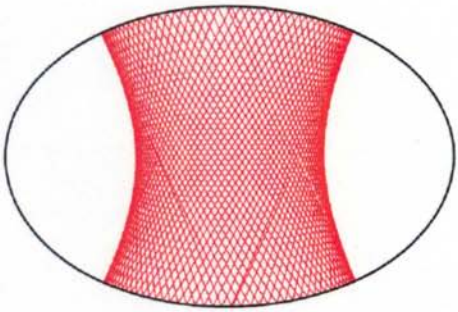
Cover Image. The image on the cover depicts an integral billiard (ellipse). This issue's article "Tea Time in Princeton" by Dr. Paul Bourgade (p. 41) discusses eigenvalues of random matrices and their connection to number theory.



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Many of us have grown up with math books and math competitions. We remember the adrenaline rush as the clock ticked down on the tests and the sleepless nights spent on attacking problems and deciphering theories. We still maintain the friendships that are made through the common interest of mathematics. That part of our childhoods have long been ingrained into our characters, and we inherit these symptoms of love for mathematics even in college.

Now we, the HCMR staff, have the honor to celebrate this love by furnishing the texts that would hopefully become the material of somebody's sleepless nights. For this process, we have had the fortune of many people's support. The two of us would like to thank the **HCMR staff** and **executive members** for the endless enthusiasm and dedication to the making of this journal. It has been a privilege to work with such committed members over the past year. The HCMR could not have grown to the level of visibility across departments around the nation and the world without your contribution.

As always, we owe our deepest gratitude to members of the **Harvard College Math Department** for the advice, encouragement and guidance throughout the publication process. We would like to thank the HCMR's **advisors** and **sponsors** for the profound contributions which continue to make this journal a success; and the **Student Organization Center at Hilles** for providing the HCMR a space in which to grow.

Finally, we would like to thank our **readers** for all the input you have given us in person and via email. Please feel free to continue to contact us with any comments, questions and concerns you may have.

This issue marks the fourth volume since the inception of the Harvard College Math Review and the second since it became an annual publication. Over the five years that the HCMR has stood as a student-run organization, it has had celebration of mathematics both in its pure and applied form at the heart of its mission. We hope you find joy in such a celebration as we did in creating this journal.

Rediet Abebe '13 and Greg Yang '14
Editors-In-Chief, The HCMR

STUDENT ARTICLE

1

A Generalization of an Action of the Braid Group on Upper-Triangular Matrices

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Abstract

We investigate various questions related to a conjecture posed by A. I. Bondal in his 2004 paper "A symplectic groupoid of triangular bilinear forms." Bondal describes an action of the braid group of n strings $\mathbf{B}_n$ on the $n \times n$ upper-triangular matrices, which has a natural (linear) algebraic formulation obtained by embedding $\mathbf{B}_n$ into a group of matrices whose entries are polynomials in $\frac{n(n-1)}{2}$ indeterminates (with an unusual composition law). One can try to extend the action to other matrices in this group to get a larger group acting on the set of upper-triangular matrices. Bondal conjectures that the action cannot enlarge past $\mathrm{Im}(\mathbf{B}_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n)$. In this paper, we prove Bondal's conjecture for the case $n = 2$, study the elements that act trivially, and present an unfinished inductive strategy for proving the conjecture for $n \geq 3$.

1.1 Introduction

Let k be an algebraically closed field of characteristic zero, let V be a vector space of dimension n over k, and let $\langle\ ,\ \rangle$ be a nondegenerate bilinear form on V. Let $\mathbb{E}$ be the set of ordered bases of V.

Definition 1. An ordered basis $E = (e_1, \ldots, e_n)$ is called *semiorthogonal* if $\langle e_i, e_j \rangle = 0$ for all $i > j$ and $\langle e_i, e_i \rangle = 1$ for all i.

Let $\mathbb{E}_s \subset \mathbb{E}$ be the set of ordered semiorthogonal bases of V and consider the group $\mathrm{Perm}(\mathbb{E}_s)$ of permutations of $\mathbb{E}_s$. Bondal proved in [1] that $\mathrm{Perm}(\mathbb{E}_s)$ contains a subgroup isomorphic to the braid group, which is defined as follows.

Definition 2. The *braid group* $\mathbf{B}_n$ has generators $\{\sigma_1, \ldots, \sigma_{n-1}\}$ with relations

$$\begin{aligned} \sigma_i\sigma_j &= \sigma_j\sigma_i && \text{if } |i-j| > 1 \\ \sigma_i\sigma_{i+1}\sigma_i &= \sigma_{i+1}\sigma_i\sigma_{i+1} && \text{for } 1 \leq i \leq n-2\,. \end{aligned}$$

Definition 3. For $i = 1, \ldots, n-1$, define the transformation $\varphi_i : \mathbb{E} \to \mathbb{E}$ which maps the ordered basis $E = (e_1, \ldots, e_n)$ to $\varphi_i(E) = (e_1', \ldots, e_n')$ where

$$\begin{aligned} e_i' &= e_{i+1} - \langle e_i, e_{i+1} \rangle e_i \\ e_{i+1}' &= e_i \\ e_j' &= e_j \text{ for } j \notin \{i, i+1\}\,. \end{aligned}$$

†Minseon Shin, Massachusetts Institute of Technology '13, is a mathematics major. His current mathematical interests include algebra and combinatorics. Outside of academics, he enjoys listening to music and playing soccer.